

Full-Text Search Engine on a Suffix-Array Index

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ABSTRACT

Full-Text Search Engine on a Suffix-Array Index is designed to provide fast and efficient searching over large collections of text documents. Unlike traditional keyword-based search methods, this system supports substring, prefix, and phrase queries, enabling more flexible and accurate information retrieval. The search engine constructs a Suffix Array in $O(n \log^2 n)$ time and computes the Longest Common Prefix (LCP) array using Kasai's Algorithm to optimize search performance. Queries are processed using binary search with LCP acceleration, reducing search time while maintaining efficient memory usage. The system also ranks search results based on the relevance and frequency of matching terms, improving the user experience. A performance comparison with a traditional inverted index highlights the advantages of the suffix-array approach for substring searching, including lower memory requirements and competitive query speed. This project demonstrates an effective and scalable solution for full-text search applications such as digital libraries, search engines, bioinformatics, and text analytics.

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